HW3

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Problem 6.1

Kinetic expression : + Single substrate (S)
+ Product (P)
+ Inhibitory effect (E-S complex can add on P for an E-S-P complex in a reversible reaction)
+ sketch (rate of product as function of substrate conc.)

Overall: $E + S \rightarrow E + P$

Michaelis-Menton: $E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES$ $\qquad ES \xrightarrow{k_2} E + P$

Product Inhibition: $ES + P \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} ESP$

@ equilibrium: $[ESP] = K_3[ES][P]$; $K_3 = \frac{k_3}{k_{-3}}$

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] = k_1[E][S] - [ES](k_{-1} + k_2)$$

$$\frac{d[P]}{dt} = k_2[ES]$$

For equilibrium reaction for step 1: $K_1 = \frac{[ES]}{[E][S]}$

$$[ES] = K_1[E][S]$$

Total conc. of enzyme sites [Site balance]:
$$[E]_0 = [E] + [ES] + [ESP]$$
$$= [E] + [ES] + K_3[ES][P]$$
$$= [E] + [ES](1 + K_3[P])$$

Substituting values for [ES],

$$[E]_0 = [E] + K_1[E][S](1 + K_3[P])$$
$$[E]_0 = [E]\{1 + K_1[S] + K_1K_3[S][P]\} \Rightarrow [E] = \frac{[E]_0}{(1 + K_1[S] + K_1K_3[S][P])}$$

Rate of product formation: $V = \frac{d[P]}{dt} = k_2[ES] - k_3[P][ES] + k_{-3}[ESP] = k_2[ES] - k_3[P][ES] + k_{-3}K_3[ES][P] = k_2[ES] + [ES][P](-k_3 + k_{-3}K_3) = k_2[ES]$ (the $-k_3 + k_{-3}K_3$ term $\rightarrow 0$)

$$\frac{d[P]}{dt} = k_2K_1[E][S] = \boxed{\frac{k_2K_1[S][E]_0}{1 + K_1[S] + K_1K_3[S][P]} = V}$$

We have to plot V vs [S]

We can find [S] as a function of [P]:
Balance on [E] and [P]: (You need 2 because one moles of S & P gives 1 mole of ESP)

$$[S]_0 = [S] + [ES] + 2[ESP] + [P]$$
$$[S]_0 = [S] + [ES] + 2(K_3[ES][P]) + [P]$$
$$[S]_0 = [S] + (K_1[E][S])(1 + 2K_3[P]) + [P]$$

Solving for [P]:
$$\boxed{[P] = \frac{[S]_0 - [S] - K_1[E][S]}{(1 + 2K_1K_3[E][S])}}$$

This is also an implicit function

Therefore combining expressions of [E], [P] and iterating [S]

Now we can iterate values for [S] and knowing constants K_1, k_2, K_3 we can plot V $[S]_0$. attached

Question 6.2

$P \overset{k_1}{\rightleftharpoons} 1P$

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Hill expression ; L = ligand ; Protein P ; $L + P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} LP$

$$\frac{d[LP]}{dt} = k_1[L][P] - 1[LP]$$

(i) Fraction of bound protein to total protein by θ:

@ Equilibrium: $\frac{1}{K} = \frac{[LP]}{[L][P]}$; $[LP] = \frac{1}{K}[L][P]$; $K = \frac{k_2}{k_1}$

(or $\frac{d[LP]}{dt} = 0$ @ss)

Total protein: $[P]_0 = [P] + [LP] = [P] + \frac{1}{K}[L][P]$

$$\theta = \frac{[LP]}{[P]_0} = \frac{\frac{1}{K}[L][P]}{[P] + \frac{1}{K}[L][P]} = \frac{\frac{1}{K}[L]}{\left(1 + \frac{1}{K}[L]\right)} = \boxed{\frac{[L]}{K + [L]} = \theta}$$

(ii) $P + 2L \xrightarrow{k_1} LP + L$; $LP \xrightarrow{k_2} L + P$

$$\frac{d[LP]}{dt} = k_1[P][L]^2 - k_2[LP]$$; Total protein: $[P]_0 = [P] + [LP]$

$$= [P] + \frac{k_1}{k_2}[P][L]^2 = [P] + \frac{1}{K}[P][L]^2$$

$$\frac{d[LP]}{dt} \approx 0 \quad - \quad [LP] = \frac{k_1}{k_2}[P][L]^2 = \frac{1}{K}[P][L]^2$$

@ss

$$\theta = \frac{[LP]}{[P]_0} = \frac{\frac{1}{K}[P][L]^2}{[P] + \frac{1}{K}[P][L]^2} = \boxed{\frac{[L]^2}{[K] + [L]^2} = \theta}$$

(iii) Based on part (ii), I would assume $P + nL \rightarrow LP + (n-1)L$; $LP \rightarrow L + P$

(iv) Scale invariance: $aA + bB \rightarrow C$; $\frac{r(\alpha[A], \beta[B])}{r([A], [B])} = \phi(\alpha, \beta)$

↓ Do not change if other variables

↑ This ratio of rates of arbitrary exponent values over the stoichiometric rate expressions

Properties of $\phi(\alpha, \beta)$: a) $\phi = 1$, when the rate expressions for stoichiometric elementary mechanisms match that of α, β expression

$r([A], [B]) = [A]^a[B]^b$

$r(\alpha[A], \beta[B]) = (\alpha[A])^a(\beta[B])^b = \alpha^a[A]^a\beta^b[B]^b = \alpha^a\beta^b[A]^a[B]^b = \alpha^a\beta^b r([A], [B])$

$$\therefore \boxed{\frac{r(\alpha[A], \beta[B])}{r([A], [B])} = \alpha^a\beta^b = f(\alpha, \beta) = \phi(\alpha, \beta)}$$

(v) $\frac{\partial r(\alpha[A], \beta[B])}{\partial \alpha} = \frac{\partial}{\partial \alpha}\left[\alpha^a\beta^b \cdot r([A], [B])\right]$; from part (iv)

$= a\alpha^{a-1}\beta^b r([A], [B])$; holding $[A]$, $[B]$, β constant

$= a\frac{\alpha^a}{\alpha}\beta^b r([A], [B])$

We also know $\phi = \alpha^a\beta^b$; $\frac{\partial \phi}{\partial \alpha} = a\alpha^{a-1}\beta^b = a\frac{\alpha^a}{\alpha}\beta^b r([A], [B])$

$$\therefore \boxed{\frac{[A]}{[A]}\frac{\partial r(\alpha[A],\beta[B])}{\partial \alpha} = r([A],[B])\cdot\frac{\partial \phi(\alpha,\beta)}{\partial \alpha}}$$

$\Rightarrow$ $x=\alpha[A]$; $y=\beta[B]$:

$$[A]\frac{\partial r(x,y)}{\partial([A]\cdot\alpha)} = r([A],[B]\cdot\beta)\frac{\partial\phi(\alpha,\beta)}{\partial\alpha} \quad ; \quad \beta=1$$

$$[A]\frac{\partial r(x,y)}{\partial x} = r([A],y)\frac{\partial\phi(\alpha,1)}{\partial\alpha}$$

$\Rightarrow$ $\alpha=1$:

$x=\alpha[A]=[A]$:

$$x\cdot\frac{\partial r(x,y)}{\partial x} = r(x,y)\frac{\partial\phi}{\partial\alpha}(1,1)$$

If we let $\mu \equiv \frac{\partial\phi(1,1)}{\partial\alpha}$

$$\Rightarrow x\frac{\partial r}{\partial x} = r\cdot\mu \rightarrow \text{This can be solved as an ODE of order 1}$$

$$\int\frac{dr}{r} = \int\frac{\mu}{x}\cdot dx$$

$$\ln r = \mu\ln x + c$$
$$\ln r = \ln x^{\mu} + c \quad ; \; c \text{ has to be a function } y \text{ since we integrated w.r.t to } x$$
$$\ln r = \ln x^{\mu} + f(y)$$

$f(y)$ can be expressed as $\ln[f(y)] = \ln[\psi(y)]$

$$\Rightarrow \ln r = \ln[\psi(y)\cdot x^{\mu}] \quad \text{Taking exponentials of both sides.}$$

$$\therefore \boxed{r(x,y) = \psi(y)x^{\mu}}$$

(vi) Following the above steps but w/ respect to β:

$$\Rightarrow \quad r(\alpha[A],\beta[B]) = r([A],[B])\cdot\alpha^a\beta^b$$

$$\Rightarrow \quad \frac{\partial r(\alpha[A],\beta[B])}{\partial\beta} = r([A],[B])\frac{\partial}{\partial\beta}(\alpha^a\beta^b)$$

$\times\frac{[B]}{[B]}$

$$[B]\frac{\partial r(\alpha[A],\beta[B])}{\partial\beta[B]} = r([A],[B])\alpha^a b\beta^b/\beta \quad ; \quad \frac{\partial\phi}{\partial\beta} = \alpha^a b\beta^b/\beta$$

$$= r([A],[B])\frac{\partial\phi(\alpha,\beta)}{\partial\beta}$$

Setting $x=\alpha[A]$, $y=\beta[B]$

$$[B]\frac{\partial r(x,y)}{\partial y} = r([A],[B])\frac{\partial\phi(\alpha,\beta)}{\partial\beta}$$

Setting $\alpha=1$, $\beta=1$:

$$y\frac{\partial r}{\partial y} = r(x,y)\frac{\partial\phi}{\partial\beta}(1,1)$$

Solving ODE:

$$y\frac{\partial r}{\partial y} = r\,\nu$$

$$\int \frac{\partial r}{r} = \int \nu \frac{\partial y}{y}$$ → since x was held constant

$$\ln r = \nu \ln y + \ln(\chi(x))$$

$$\boxed{r(x,y) = \chi(x)\, y^{\nu}}$$

(vii) Hill expression: $L + P \rightarrow LP$
[A] [B] [C]

As seen from above expressions, values of a & b as stoichiometric coefficients are absorbed by ϕ term and the final rate expressions have similar expressions regardless of the value of a & b proposed. Hence those can be non-integer.

(i.e. α & β can scale accordingly to obtain $r(x,y) = \lambda x^{\mu} y^{\nu}$ where $\mu = \frac{\alpha^a \beta^b}{\alpha}$; $\nu = \frac{\alpha^a \beta^b}{b}$ can be any non-integer value)